MATH 243 — Probability and Statistics

Discrete Distributions: Bernoulli, Binomial, Poisson, Multinomial, Hypergeometric

Lesson notes · Exam approach · TF practice with solutions · Quick notes (KKS Sir's portion — Probability & ANOVA)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

Contents

Exam Map — What the TF Papers Ask — binomial and Poisson in almost every paper
Lesson 13 — Bernoulli and Binomial Distributions — pmf, proofs of mean and variance, numericals
Lesson 14 — Poisson Distribution and the Poisson Approximation — pmf, mean = variance, limit proof, approximations
Lesson 15 — Multinomial, Hypergeometric and Geometric Distributions — other distributions + TF questions
Quick Notes — Discrete Distributions — revision bullets

NOTE Page numbers refer to KKS Sir's Lecture 10.pdf (4 pp.), Lecture 11.pdf (5 pp.) and Lecture on other distributions...pdf (Walpole 9th ed. extracts: multinomial pp. 3–4, exercises pp. 4–6, hypergeometric introduction p. 6). The hypergeometric pmf and the geometric/negative-binomial distributions are completed from the textbook and marked. "TF p." = merged question PDF page; "22 batch p." = 2023-24 paper.

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note [lesson]
2023-24 (MATH 243) TF p. 22 batch p.2	Q3(b) Relation between binomial and Poisson; 0.5 calls/min: P(2 calls in 5 minutes) (12) [L14] · Q3(c) P(recover) 0.4, 15 patients: P(at least 10 survive), P(3 to 8 survive), P(exactly 5 die) (12) [L13]
2022-23 (MATH 243) TF p. 48	Q3(a) Bernoulli and Poisson processes; derive the Poisson distribution (10) [L14] · Q3(b) Variance of Poisson; 1 in 1000 returns has an error, 10,000 examined: P(6, 7 or 8 errors) (12) [L14]
2021-22 (MATH 245) TF p. 44	Q6(a) 500 lottery tickets, 200 pay off; buy 5: P(win back at least the cost of 3) (10) [L15] · Q6(b) Delegates by air/bus/auto/train (0.4, 0.2, 0.3, 0.1): P(3, 3, 1, 2 among 9) (10) [L15] · Q6(c) 5 cars, 2 blemished; agency receives 3: probability distribution and F(x) of the number blemished (15) [L15]
2020-21 (MATH 245) TF p. 38	Q6(c) Valium, 2/3 of users women: 5th prescription is the first / the third for a woman (10) [L15] · Q7(a) Diode failure 0.03, 200 diodes: mean, variance, P(board works = no failures) (17) [L13]
2019-20 (MATH 245) TF p. 35	Q8(a) Bubbles in glass, 1 in 1000; sample of 8000: P(fewer than 7) (Poisson approximation) (10) [L14]
2016-17 (MATH 245) TF p. 21	Q8(a) Prove $b(x; n, p) \to$ Poisson as $n \to \infty$, $p \to 0$; accidents $p = 0.005$ over 400 days: P(exactly 1), P(at most 3) (20) [L14]
2014-15 (MATH 241) TF p. 11	Q8(b) P(graduate) = 0.4; of 5 students: P(none), P(at least 1), P(all) (12) [L13]
2013-14 (MATH 241) TF p. 8	Q7(b)(ii) 10 % defective, 12 instruments: P(at least 4 defective) (8) [L13]
2012-13 (MATH 241) TF p. 5	Q6(b) Truck tyres, 20 % blow out, next 18: P(4 to 7), P(fewer than 3), P(more than 5) by binomial and Poisson (10) [L14] · Q6(c) Derive the mean and variance of the binomial distribution (10) [L13]
2011-12 (MATH 241) TF p. 2	Q7(b)(i) Insurance: P(death) = 0.0005, 1000 policies — P(5 or more claims) (7) [L14]

EXAM FOCUS 16 questions, 185 marks — the most frequently asked probability topic (10 questions since 2019-20). Three things are certain to come: (1) a binomial numerical with "at least / from a to b"; (2) a Poisson numerical, usually as an approximation to the binomial with $\lambda = np$; (3) a derivation — mean and variance of the binomial, or the Poisson as a limit of the binomial (2012-13, 2016-17, 2022-23, 2023-24). Learn the two derivations word-perfect.

Lesson 13 — Bernoulli and Binomial Distributions

Lec 10 pp. 1–4

1. Binomial Experiment (Lec 10 p. 1)

An experiment with two outcomes (success, failure), repeated n times independently, with the same probability p of success in each trial, is a binomial experiment. X = number of successes.

$$X \sim \text{Binomial}(n, p):\ P(X = x) = {}^{n}C_{x}\, p^{x} (1-p)^{n-x},\ x = 0, 1, \ldots, n$$
$$E(X) = np,\quad V(X) = np(1-p)$$

n = 1 gives the **Bernoulli** distribution: $P(X = x) = p^{x}(1-p)^{1-x}$, x = 0, 1, with E(X) = p, V(X) = p(1 − p).

2. The Proofs (Lec 10 p. 1 asks them; standard working) ★

```
Bernoulli:  Σ f = (1 − p) + p = 1;  E(X) = 0(1 − p) + 1·p = p;  E(X²) = p;  V(X) = p − p² = p(1 − p)

Binomial total:  Σ_{x=0}^{n} nCx p^x q^(n−x) = (p + q)^n = 1^n = 1      (q = 1 − p, binomial theorem)

Mean:  E(X) = Σ x · n!/(x!(n−x)!) p^x q^(n−x)          (the x = 0 term is 0)
           = np Σ_{x=1}^{n} (n−1)!/((x−1)!(n−x)!) p^(x−1) q^(n−x)
           = np Σ_{y=0}^{n−1} (n−1)Cy p^y q^(n−1−y)    (y = x − 1)
           = np (p + q)^(n−1) = np

E[X(X−1)] = Σ x(x−1) n!/(x!(n−x)!) p^x q^(n−x)
          = n(n−1)p² Σ_{y=0}^{n−2} (n−2)Cy p^y q^(n−2−y) = n(n−1)p²
E(X²) = E[X(X−1)] + E(X) = n(n−1)p² + np
V(X) = E(X²) − (np)² = n²p² − np² + np − n²p² = np(1 − p) = npq
```

MEMORIZE Trick for both binomial and Poisson variances: find E[X(X − 1)] first (the x(x − 1) cancels with x!), then V = E[X(X − 1)] + E(X) − [E(X)]².

3. Worked Examples (Lec 10 pp. 2–3)

```
Recover p = 0.9, n = 7:  P(X = 5) = 7C5 (0.9)^5 (0.1)^2 = 0.124;   E(X) = 7 × 0.9 = 6.3 patients
Trucks out of city p = 0.25, n = 5:
   P(X ≥ 3) = 5C3(.25)³(.75)² + 5C4(.25)⁴(.75) + (.25)⁵ = 0.1035
   P(X > 3) = P(4) + P(5) = 0.0156;   E(X) = 5 × 0.25 = 1.25 (125 of 500 trucks)
```

- Interval words: at most 2 → X ≤ 2; less than 2 → X ≤ 1; P(1 ≤ X < 3) = P(1) + P(2); P(1 < X < 3) = P(2).
- Given E(X) = 5 and V(X) = 2: 1 − p = 2/5, p = 3/5, n = 5/p = 25/3. Since n must be an integer, such a binomial cannot exist — the slide stops at n = 25/3; say so in an answer.

Exercises of Lecture 10 (p. 4) — answers

Exercise	Answer
10 tosses, exactly 7 heads	$10C7/2^{10}$ = 120/1024 = 0.1172
20 items, p = 0.05: at most 1; at least 3; between 2 and 5	0.7358; 0.0755; P(2 ≤ X ≤ 5) = 0.2638
15 patients, p = 0.6: at least 10; mean, variance	0.4032; 9 and 3.6

Exam Approach

1. Define X and the success, check the four binomial conditions, write X ~ Bin(n, p).
2. Translate the words into an inequality on X.
3. Use the complement when the requested side has more terms ("at least 1" = 1 − P(0)).
4. Write every term $nCx\ p^x q^{n-x}$ explicitly, then the sum; keep 4 decimals.

Practice — Lesson 13

(a) Actual — 2023-24 Q3(c) (22 batch p. 2). (b) Actual — 2014-15 Q8(b) (TF p. 11). (c) Actual — 2013-14 Q7(b)(ii) (TF p. 8). (d) Actual — 2020-21 Q7(a) (TF p. 38). (e) Actual — 2012-13 Q6(c) (TF p. 5).

(a) The probability that a person recovers from a rare blood disease is 0.4. If 15 people are known to have contracted it, find the probability that (i) at least 10 survive, (ii) from 3 to 8 survive, (iii) exactly 5 die. (12)

(b) The probability that an entering college student will graduate is 0.4. Out of 5 students find the probability that (i) none, (ii) at least 1, (iii) all will graduate. (12)

(c) 10 % of the instruments produced are defective. Find the probability that out of 12, at least 4 are defective. (8)

(d) The probability of a diode failure is 0.03 and a board contains 200 diodes. (i) Mean number of failures? (ii) Variance? (iii) The board works only if no diode is defective: P(board works)? (17)

(e) Derive the mean and variance of the binomial distribution. (10)

✓ Solution

(a) X = number who survive (recover), X ~ Bin(15, 0.4), q = 0.6.

```
(i)   P(X ≥ 10) = Σ_{x=10}^{15} 15Cx (0.4)^x (0.6)^(15−x)
            = 0.0245 + 0.0074 + 0.0016 + 0.0003 + 0 + 0 = 0.0338
(ii)  P(3 ≤ X ≤ 8) = 0.0634 + 0.1268 + 0.1859 + 0.2066 + 0.1771 + 0.1181 = 0.8778
(iii) 5 die ⇔ 10 survive:  P(X = 10) = 15C10 (0.4)^10 (0.6)^5 = 3003 × 1.048576e−4 × 0.07776 =
0.0245
```

(b) X ~ Bin(5, 0.4).

```
(i) P(0) = 0.6^5 = 0.0778   (ii) P(X ≥ 1) = 1 − 0.6^5 = 0.9222   (iii) P(5) = 0.4^5 = 0.0102
```

(c) X ~ Bin(12, 0.1): P(X ≥ 4) = 1 − [P(0) + P(1) + P(2) + P(3)].

```
P(0) = 0.9^12 = 0.2824;  P(1) = 12(0.1)(0.9)^11 = 0.3766;  P(2) = 66(0.01)(0.9)^10 = 0.2301;
P(3) = 220(0.001)(0.9)^9 = 0.0852
P(X ≥ 4) = 1 − 0.9744 = 0.0256
```

(d) X = failures ~ Bin(200, 0.03).

```
(i)   E(X) = np = 200 × 0.03 = 6
(ii)  V(X) = np(1 − p) = 6 × 0.97 = 5.82
(iii) P(X = 0) = 0.97^200 = 0.0023   (Poisson λ = 6: e^−6 = 0.0025, close)
```

(e) The derivation in §2 (total = 1, E(X) = np via the y = x − 1 shift, V(X) = npq via E[X(X − 1)]).

Lesson 14 — Poisson Distribution and the Poisson Approximation

Lec 11 pp. 1–5

1. The Poisson Distribution (Lec 11 p. 1)

$$X \sim \text{Poisson}(\lambda):\ f(x) = e^{-\lambda}\lambda^x / x!,\ x = 0, 1, 2, \ldots$$
$$E(X) = V(X) = \lambda$$

λ = mean number of successes in a given time interval or region; X = number of successes in it. Used for counts of rare events (calls per interval, accidents per week, defects per item).

2. Proof that Mean = Variance (Lec 11 p. 1) ★

```
Σ f(x) = e^-λ Σ λ^x/x! = e^-λ e^λ = 1
E(X) = Σ_{x=1}^∞ x e^-λ λ^x/x! = λ e^-λ Σ_{x=1}^∞ λ^(x-1)/(x-1)! = λ e^-λ e^λ = λ
E[X(X-1)] = Σ_{x=2}^∞ x(x-1) e^-λ λ^x/x! = λ² e^-λ Σ_{x=2}^∞ λ^(x-2)/(x-2)! = λ²
V(X) = E[X(X-1)] + E(X) − [E(X)]² = λ² + λ − λ² = λ
```

3. Poisson as a Limit of the Binomial ★ (Lec 11 p. 3 states it; proof: textbook)

```
Let n → ∞, p → 0 with np = λ fixed, i.e. p = λ/n.
b(x; n, p) = n!/(x!(n-x)!) (λ/n)^x (1 − λ/n)^(n-x)
           = (λ^x/x!) · [n(n-1)…(n-x+1)/n^x] · (1 − λ/n)^n · (1 − λ/n)^(-x)
As n → ∞:  n(n-1)…(n-x+1)/n^x = (1)(1 − 1/n)…(1 − (x-1)/n) → 1
           (1 − λ/n)^n → e^-λ,      (1 − λ/n)^(-x) → 1
Hence b(x; n, p) → e^-λ λ^x / x! = p(x; λ)   ■
```

EXAM-READY ANSWER Bernoulli process: a sequence of independent trials, each a success with the same probability p; the number of successes in n trials is binomial. Poisson process: events occur randomly in continuous time (or space) such that (i) counts in non-overlapping intervals are independent, (ii) the probability of one event in a short interval of length h is about λh, proportional to h, and (iii) the probability of two or more events in it is negligible. Dividing an interval of length t into n small sub-intervals turns the Poisson process into a Bernoulli process with p = λt/n; letting n → ∞ gives the Poisson distribution with mean λt (the limit above). Relation: Poisson(np) is the limit of Binomial(n, p) when n is large and p small. (2022-23 Q3(a), 2023-24 Q3(b), 2016-17 Q8(a))

4. Worked Examples (Lec 11 pp. 1–4)

```
Calls, λ = 3 per 10 min:  P(0) = e^-3 = 0.0498;  P(1) = 3e^-3 = 0.1494;  P(X ≥ 2) = 1 − 0.0498
− 0.1494 = 0.8009
P(X = 1) = P(X = 2):  e^-λ λ = e^-λ λ²/2  →  λ = 2;  mean 2, SD √2
Beakers n = 250, p = 0.012:  binomial P(2) = 0.2245;  Poisson λ = 3: P(2) = 9e^-3/2 = 0.224
Accidents p = 0.005, 400 days (λ = 2):  P(1) = 2e^-2 = 0.2707;  P(X ≤ 3) = 0.8571
Bubbles 1 in 1000, n = 8000 (λ = 8):  P(X < 7) = Σ_0^6 e^-8 8^x/x! = 0.3134
```

Exercises of Lecture 11 (pp. 2–5) — answers

Exercise	Answers
λ = 4 per hour: P(7), P(0)	P(7) = 0.0595; P(0) = 0.0183
λ = 2 per week: P(≤ 1), P(≥ 2)	≤ 1 = 0.406; ≥ 2 = 0.594

Exercise	Answers
λ = 6 per hour: P(4 to 8), P(> 10)	4–8 = 0.696; > 10 = 0.0426
1000 × 0.002 (λ = 2): P(3), P(≤ 3), P(≥ 3)	3 = 0.1804; ≤ 3 = 0.8571; ≥ 3 = 0.3233
500 × 0.01 (λ = 5): P(4), P(≤ 4), P(≥ 6)	4 = 0.1755; ≤ 4 = 0.4405; ≥ 6 = 0.384
600 × 0.005 (λ = 3): P(2), P(≤ 2), P(≥ 4)	2 = 0.224; ≤ 2 = 0.4232; ≥ 4 = 0.3528
2000 × 0.001 (λ = 2): P(3), P(≤ 3), P(≥ 3)	3 = 0.1804; ≤ 3 = 0.8571; ≥ 3 = 0.3233
8000 × 0.0005 (λ = 4): P(2), P(≤ 2), P(≥ 3)	2 = 0.1465; ≤ 2 = 0.2381; ≥ 3 = 0.7619

TRAP Scale λ to the interval asked: 0.5 calls per minute → λ = 2.5 for a five-minute period. And use the Poisson approximation only when n is large and p small (np moderate); otherwise compute the binomial.

Exam Approach

1. Identify the count and its mean λ for the stated interval (λ = np for an approximation).
2. Write $f(x) = e^{-\lambda}\lambda^x/x!$ for each x needed; use 1 − Σ for "at least".
3. When asked for both binomial and Poisson, compute both and comment on how close they are.
4. Derivations: Σ f = 1, E(X) by cancelling x with x!, V(X) via E[X(X − 1)]; limit proof with p = λ/n.

Practice — Lesson 14

(a) Actual — 2022-23 Q3(b) (TF p. 48). (b) Actual — 2023-24 Q3(b) (22 batch p. 2). (c) Actual — 2016-17 Q8(a) (TF p. 21). (d) Actual — 2019-20 Q8(a) (TF p. 35). (e) Actual — 2011-12 Q7(b)(i) (TF p. 2). (f) Actual — 2012-13 Q6(b) (TF p. 5).

(a) Calculate the variance of a Poisson random variable. On average 1 person in 1000 makes a numerical error in his or her tax return. If 10,000 returns are examined, find the probability that 6, 7 or 8 contain an error. (12)

(b) Establish the relation between the binomial and Poisson distributions. Calls arrive at a mean rate of 0.5 per minute. Find the probability that two calls arrive in a particular five-minute period. (12)

(c) Prove that b(x; n, p) → p(x; μ) when n → ∞, p → 0 and np → μ. Accidents occur on a day with probability 0.005, independently. Find the probability that in 400 days (i) there is an accident on one day, (ii) there are at most three days with an accident. (20)

(d) On average 1 in every 1000 glass items has bubbles. What is the probability that a random sample of 8000 yields fewer than 7 items with bubbles? (10)

(e) P(an insured person dies within a year) = 0.0005. If 1000 such persons are insured, find the probability that the company receives 5 or more claims in a year. (7)

(f) 20 % of trucks fail the test run with a blowout. Of the next 18 trucks, find the probability, by both the binomial and the Poisson distribution, that (i) from 4 to 7, (ii) fewer than 3, (iii) more than 5 have blowouts. (10)

✓ Solution

(a) Variance = λ (proof in §2). Here λ = np = 10,000 × 0.001 = 10.

```
P(6) = e^-10 10^6/6! = 0.0631;  P(7) = e^-10 10^7/7! = 0.0901;  P(8) = e^-10 10^8/8! = 0.1126
P(6 or 7 or 8) = 0.2657
```

(b) Relation: §3 (the Poisson is the limit of the binomial with λ = np). For five minutes λ = 0.5 × 5 = 2.5.

$$\mathbf{P(X = 2) = e^{-2.5}(2.5)^2/2! = 3.125 \times 0.082085 = 0.2565}$$

(c) Proof in §3. λ = np = 400 × 0.005 = 2.

```
(i)  P(X = 1) = 2e^−2 = 0.2707
(ii) P(X ≤ 3) = e^−2 (1 + 2 + 2²/2! + 2³/3!) = e^−2 (1 + 2 + 2 + 1.3333) = 6.3333 × 0.135335 = 0.8571
```

(d) n = 8000, p = 0.001 → λ = 8.

```
P(X < 7) = e^−8 [1 + 8 + 8²/2! + 8³/3! + 8⁴/4! + 8⁵/5! + 8⁶/6!]
         = 0.00033546 × [1 + 8 + 32 + 85.333 + 170.667 + 273.067 + 364.089]
         = 0.00033546 × 934.156 = 0.3134
```

(e) λ = 1000 × 0.0005 = 0.5.

```
P(X ≥ 5) = 1 − e^−0.5 [1 + 0.5 + 0.125 + 0.020833 + 0.0026042] = 1 − 0.606531 × 1.6484375 = 0.000172
(exact binomial: 0.000171; about 1.7 chances in 10,000)
```

(f) Binomial X ~ Bin(18, 0.2); Poisson λ = np = 3.6.

Part	Binomial	Poisson (λ = 3.6)
(i) 4 ≤ X ≤ 7	0.4827	0.454
(ii) X ≤ 2	0.2713	0.3027
(iii) X ≥ 6	0.1329	0.1559

Individual binomial terms: P(0) = 0.0180, P(1) = 0.0811, P(2) = 0.1723, P(4) = 0.2153, P(5) = 0.1507, P(6) = 0.0816, P(7) = 0.0350. The Poisson values differ by 0.02–0.03 because p = 0.2 is not small.

Lesson 15 — Multinomial, Hypergeometric and Geometric Distributions

other distributions pp. 3–6 · hypergeometric pmf and geometric: textbook

1. Multinomial Distribution (other distributions pp. 3–4)

Each of n independent trials ends in one of k outcomes $E_1, \ldots, E_k$ with probabilities $p_1, \ldots, p_k$ ($\Sigma p_i = 1$). The number of orders giving x_1 of $E_1, \ldots, x_k$ of E_k is $n!/(x_1!\ldots x_k!)$, each with probability $\Pi\, p_i^{x_{\{i}}}$.

$$f(x_1, \ldots, x_k) = [n! / (x_1!\, x_2! \ldots x_k!)]\; p_1^{x_{\{1}}}\, p_2^{x_{\{2}}} \ldots p_k^{x_{\{k}}},\quad \Sigma x_i = n$$

Example 5.7 (runways 2/9, 1/6, 11/18; 6 planes as 2, 1, 3): $6!/(2!1!3!)\,(2/9)^2(1/6)(11/18)^3 = 60 \times 0.04938 \times 0.16667 \times 0.22822 = 0.1127$.

2. Hypergeometric Distribution (other distributions p. 6; pmf: textbook)

Sampling **without replacement** from a lot of N items of which k are successes (defectives, winning tickets): trials are not independent, so the binomial does not apply exactly.

$$h(x;\, N, n, k) = kCx \cdot (N - k)C(n - x) \,/\, NCn$$
$$E(X) = nk/N, \quad V(X) = [(N - n)/(N - 1)] \cdot n\,(k/N)(1 - k/N)$$

When N is large compared with n, $h(x; N, n, k) \approx b(x; n, k/N)$.

3. Geometric and Negative Binomial (textbook)

$$\text{Geometric (first success on trial x): } g(x; p) = p\, q^{x-1},\quad x = 1, 2, \ldots$$
$$\text{Negative binomial (k-th success on trial x): } b^*(x; k, p) = (x-1)C(k-1)\, p^k\, q^{x-k}$$

Exam Approach

1. Classify: fixed n with replacement/independent → binomial; without replacement from a finite lot → hypergeometric; more than two outcomes → multinomial; "the x-th trial is the first/k-th success" → geometric/negative binomial.
2. Write the formula with the numbers substituted before multiplying.
3. Distribution + F(x): list f(x) for every x, then cumulative sums as a step function.

Practice — Lesson 15

(a) Actual — 2021-22 Q6(b) (TF p. 44). (b) Actual — 2021-22 Q6(a) (TF p. 44). (c) Actual — 2021-22 Q6(c) (TF p. 44). (d) Actual — 2020-21 Q6(c) (TF p. 38).

(a) Delegates arrive by air, bus, automobile or train with probabilities 0.4, 0.2, 0.3, 0.1. What is the probability that among 9 delegates 3 arrived by air, 3 by bus, 1 by automobile and 2 by train? (10)

(b) 500 lottery tickets are sold; 200 of them pay off at least the cost of the ticket. You buy 5 tickets. Find the probability that you win back at least the cost of 3 tickets. (10)

(c) A shipment of 5 cars contains 2 with slight paint blemishes. An agency receives 3 at random. Find the probability distribution and F(x) of X = number of blemished cars received. (15)

(d) Two-thirds of Valium users are women. Find the probability that on a given day the fifth prescription written by a doctor is (i) the first prescribing Valium for a woman, (ii) the third prescribing Valium for a woman. (10)

✓ Solution

(a) Multinomial with n = 9.

$$\mathbf{f = 9!/(3!\,3!\,1!\,2!)\,(0.4)^3(0.2)^3(0.3)(0.1)^2 = 5040 \times 0.064 \times 0.008 \times 0.3 \times 0.01 = 0.00774}$$

(b) X = winning tickets among 5; N = 500, k = 200, n = 5 (without replacement).

```
P(X ≥ 3) = [200C3·300C2 + 200C4·300C1 + 200C5·300C0] / 500C5
         = 0.2308 + 0.076 + 0.0099 = 0.3167
Binomial check (N large, p = 200/500 = 0.4): P(X ≥ 3) = 0.3174 — almost the same
```

(c) Hypergeometric N = 5, k = 2, n = 3; 5C3 = 10. X can be 0, 1, 2.

```
f(0) = 2C0·3C3/10 = 1/10;   f(1) = 2C1·3C2/10 = 6/10 = 3/5;   f(2) = 2C2·3C1/10 = 3/10
F(x) = 0 for x < 0;  1/10 for 0 ≤ x < 1;  7/10 for 1 ≤ x < 2;  1 for x ≥ 2
```

(d) Each prescription is for a woman with p = 2/3, independently (q = 1/3).

```
(i)  geometric: P(first woman on 5th) = q⁴ p = (1/3)⁴(2/3) = 2/243 = 0.0082
(ii) negative binomial, k = 3, x = 5: 4C2 (2/3)³(1/3)² = 6 × 8/27 × 1/9 = 16/81 = 0.1975
```

Quick Notes — Discrete Distributions

Revision points for Lessons 13–15.

L13 · Binomial

- nCx $p^x q^{n-x}$; mean np; variance npq (prove via y = x − 1 and E[X(X − 1)]).
- Bernoulli: p, p(1 − p). E = 5, V = 2 gives n = 25/3 — impossible binomial.
- Bin(15, 0.4): P(≥ 10) = 0.0338, P(3–8) = 0.8778, P(10) = 0.0245. Bin(12, 0.1): P(≥ 4) = 0.0256.
- Diodes: mean 6, var 5.82, $P(0) = 0.97^{200} = 0.00226$.

L14 · Poisson

- $e^{-\lambda}\lambda^x/x!$; mean = variance = λ; limit of Bin(n, λ/n) as $n \to \infty$.
- λ = np when n large, p small; scale λ to the interval (0.5/min × 5 = 2.5).
- Tax returns λ = 10: 0.2657. Accidents λ = 2: 0.2707, 0.8571. Bubbles λ = 8: 0.3134. Claims λ = 0.5: 0.00017.

L15 · Other Discrete

- Multinomial $n!/(\Pi x_i!)\ \Pi p_i^{x_i}$; delegates 0.00774.
- Hypergeometric kCx(N−k)C(n−x)/NCn, mean nk/N; lottery 0.3167; cars 1/10, 6/10, 3/10.
- Geometric pq^{x-1}; negative binomial $(x-1)C(k-1)p^k q^{x-k}$; Valium 2/243, 16/81.